Pipeline maps — as-built and as-designed

Metrics quoted in this document

Unless a figure says otherwise it is per-stream nRMSE — RMSE normalised by that stream's range, then averaged — which was this programme's headline metric until the metric error recorded in `EKF_MHT_DESIGN.md` §6. Normalising per stream lets a stream with a tiny range outvote the largest one, which is how a correct mechanism got recorded as a failure. **Pooled RMSE in deaths** replaced it as the headline and is always named explicitly here. The two are not comparable: 0.591 is dimensionless, 29.3 is a death count.

metric	units	where it appears
nRMSE	dimensionless	the default below; every pre-2026-08-24 figure
pooled RMSE	deaths	the global-decode row, and anywhere it says "deaths"
binding precision	%	of figures bound to a role, the share that are correct
strict F1	—	exact span *and* role match

Thresholds and knobs (`--gate-threshold 0.5`, ratio 2.0, σ=0.3) are settings, not measurements.

Two pipelines, drawn separately because they are two programmes that share an extractor, not two halves of one:

	unit	span	ends when
standard event pipeline	one document	within-document	the document ends
EKF pipeline	one real-world event	across documents and across time	the event stops being reported

The boundary between them is a single record type — the **observation** `(t, role, value, qualifier, source, event_key)`. Everything upstream produces one; everything downstream consumes one. That seam is also where every hard problem of the last week lives: attachment, scope, and cross-event contamination are all failures to build a correct observation, not failures of either pipeline's internals.

✅ shipped and measured ⚠ shipped with a known defect ⛔ specified, not built

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1. Standard event pipeline — AS BUILT

`gliner2/` — one document in, structured records out. No time, no tracking, no state.

```
text (str)  +  Schema{entities, relations, json_structures, events, classifications}
      |
      v
[A] SchemaTransformer                                        processor.py
      |   schema dict -> per-task schema token sequences
      |   markers: [P] prompt  [E] entity  [C] class  [R] relation  [V] value  [L] label
      |   record_metadata declares mode: natural | latent | anchorless
      v
[B] ExtractorCollator -> PreprocessedBatch                   training/trainer.py
      |   tokenize, pad, build start/end word->token maps
      v
[C] BaseExtractorModel._encode_core                          models/boundary/model.py
      |   encoder (DeBERTa-v3 / mmBERT) -> text_states, query_states
      |   ext_specs: one query per (task, field);  QueryLayout types them by role_name
      v
[D] BoundaryHead -> CandidateTensorBatch                     models/boundary/heads.py
      |   start/end boundary scores -> sparse (query, candidate) spans (start/end_top_k=128)
      |   pair_split_reduction: true, hard_negative_keep_all_when_absent: false (eb22g)
      |   NOT fixed-width enumeration: removes the span 19-instance cap
      v
      +---------------------------+---------------------------+
      |                           |                           |
      v decode_mode=greedy (ship)  v decode_mode=joint (parity) v
[E1] per-query + record decode [E2] joint_ie                 [E3] classifications
     _decode_records (records.py):  boundary_candidates_to_       pooled logits
     - event_records: true          candidate_score_set           -> labels
     - record_link_mode: junction   + relation pairs -> edges
       (gated bilinear + geom)      + record groups -> role edges
     - TriggerLink coref merge      -> JointProblem
       (coref_link_distance: mlp)   -> BeamOptimizer (W=1)
     - EntityCorefLink clusters
     _decode_relations (biaffine)
     null-abstention ✅
      |                           |
      +---------------------------+
      v
[F] span -> char offsets, format_results                     inference/engine.py
      |   field confidence = assignment probability (pair_confidence stored alongside)
      v
 {entities: [...], relations: [...], event_extraction: {...}, structures: {...}, classifications: {...}}
```

Long documents bypass nothing — they re-enter at [A] per chunk:

```
document --> chunk (words, overlapping) --> [A..F] per chunk --> merge by char offset
            extract_long / batch_extract_long, chunk_size + chunk_overlap
```

As-built notes that matter when working in this code

- **[E2] beam width should be 1.** Relation F1 falls monotonically 0.2406 (W=1) to 0.2058 (W=64); entity metrics do not move at all, because `_finish_nodes` admits every positive-score node regardless of beam state. Default is still 16. *(JOINT_IE_SCALING §4c)*

- **[E2] gained the eval threshold on 2026-08-10.** Before that, `decision_threshold` was pinned at 0.5 and the joint decode ignored `--threshold` for edge selection entirely.
- ⚠ **[E1] vs [E2] are not threshold-identical.** Null-abstention and adaptive thresholding are greedy-only. Any arm comparison inherits that asymmetry.
- **Record role edges bypass threshold centering** — a scalar role's utility is the ABSENT-relative `logit_c − logit_ABSENT`, which has no probability cutoff to centre on.
- ⚠ **`json_structures` without `record_metadata`:** All corpora on disk were repaired on 2026-09-08 (`stamp_record_metadata.py`, gated by `audit_corpora.py`), and both the viewer and `tools/infer.py` (`_declare_records`) now default omitted structure metadata to `mode="natural", anchor="<first field>"`.
- ✅ **Events now train and decode on the record head (`eb17` → `eb22g`, `PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`).** With `event_records: true`, each trigger candidate seeds a distinct event instance; `record_link_mode: junction` binds arguments to triggers via a role-gated bilinear + geometry join; `TriggerLink` (`record_coref_link: true`, 64-width MLP distance) merges coreferent trigger mentions into document-level event clusters; and `eb22g` (`pair_split_reduction: true`, `hard_negative_keep_all_when_absent: false`, split `_list_field_bce`) resolves both role-assignment and candidate-proposal gradient starvation.

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2. EKF pipeline — AS BUILT

`tools/ekf_showcase/run_pipeline.py` — a feed of articles in, a tracked state over time out. Stage 2 calls the *whole* of pipeline 1 as a subroutine.

```
feed.jsonl   [{t_hours, text, url}, ...]        one row per archived article
      |
      v
[0a] LANGUAGE GATE              language_gate.py, SUPPORTED={en,zh,tr}       ✅
      |   rejects unsupported scripts/languages before the neural gate runs
      v
[0] GATE                        classification, gliner2-gate2-mmbert-tr      ✅
      |   "is this article a mass-casualty report?"  --gate-threshold 0.9
      |   ⚠ answers "is this article ABOUT one", never "does this number
      |     belong to THAT one" -- the source of cross-event contamination
      v
[1] EVENT (optional)            DocEE classification, --event-model          ✅
      |   event_type + 'Casualties and Losses' spans
      |   SKIPPED when --event-model is unset; then event_type = 'unknown'
      |   and --associate envelope / type+location degrade to no key
      v
[2] EXTRACT                     casualty structure model  === PIPELINE 1 === ✅
      |   default: whr778/gliner2-casualty-multilingual (to be rebased onto eb22g)
      |   Schema().structure("casualty_report", mode="natural", anchor="dead")
      |     .field(dead|injured|missing, exclusive=True) .field(location)
      |   --record-mode natural   (REQUIRED for boundary models)
      |   --window article | event | lead | long   (long = extract_long)
      v
[3] NORMALIZE                   span -> (value, qualifier, source)           ✅
      |   --normalizer heuristic | classify | hybrid | both
      |   word_number() parses spelled-out cardinals; unparsable -> None,
      |     never 0 (that bug fabricated 30 zeros)
      |   qualifier: point|at_least|about|feared|interval   source: official|...
      v
[3b] TEMPORAL FILTER            --event-year                                 ✅
      |   drop figures dated before the event: Izmit 1999's 17,500 bound as
      |   a 2023 observation 15 times -> 3, with zero genuine losses
      v
[3c] ASSOCIATE                  --associate none|type|type+location|envelope|record
      |   record   : location from the record's OWN field   (strongest signal)
      |   envelope : nearest location span by char distance (heuristic)
      |   -> event_key,  e.g. "Floods|north carolina"
      v
[3d] ROLLUP & SCOPE FILTER      --rollup datasets/<event>/rollup.json        ✅
      |   collapse_type: Floods|florida == Storm|florida
      |   aliases:   asheville -> north carolina;  six states -> __aggregate__
      |   hierarchy: {aggregate: __aggregate__, parts: [...]}   <- declares scope
      |   --scope-filter: drops places outside declared hierarchy (4/6 cross-event @ 1.2% FP)
      v
[3e] GLOBAL SCOPE DECODE (M4')  --scope-method decode (scope_gate.hmm_gate)  ✅ DEFAULT
      |   Viterbi decode over {own, aggregate, reject} across the full stream
      |   (sigma=0.3, reject_cost=4.0, stay=0.1, warmup=0; legacy ratio=2.0 available)
      |   pooled RMSE in deaths: Helene 29.3 -> 20.7 (-29.4%),
      |     Turkiye 11,581.5 -> 10,695.5 (-7.6%), Aegean 74.4 -> 15.7 (-78.8%)
      v
[4] TRACK                       per event_key                                ✅
      |   HARD assignment on the string key -> one single-stream EKF per key
      |   est_ekf(observations, grid, role)   vs   est_last_value (baseline)
      v
 tracked.json  {grid, tracked_by_event: {key: {role: {ekf, last_value, n_obs}}}}
```

```
        |
        v
scoring: score_helene.py | revision_test.py | vector_state_test.py | scope_gate_test.py
```

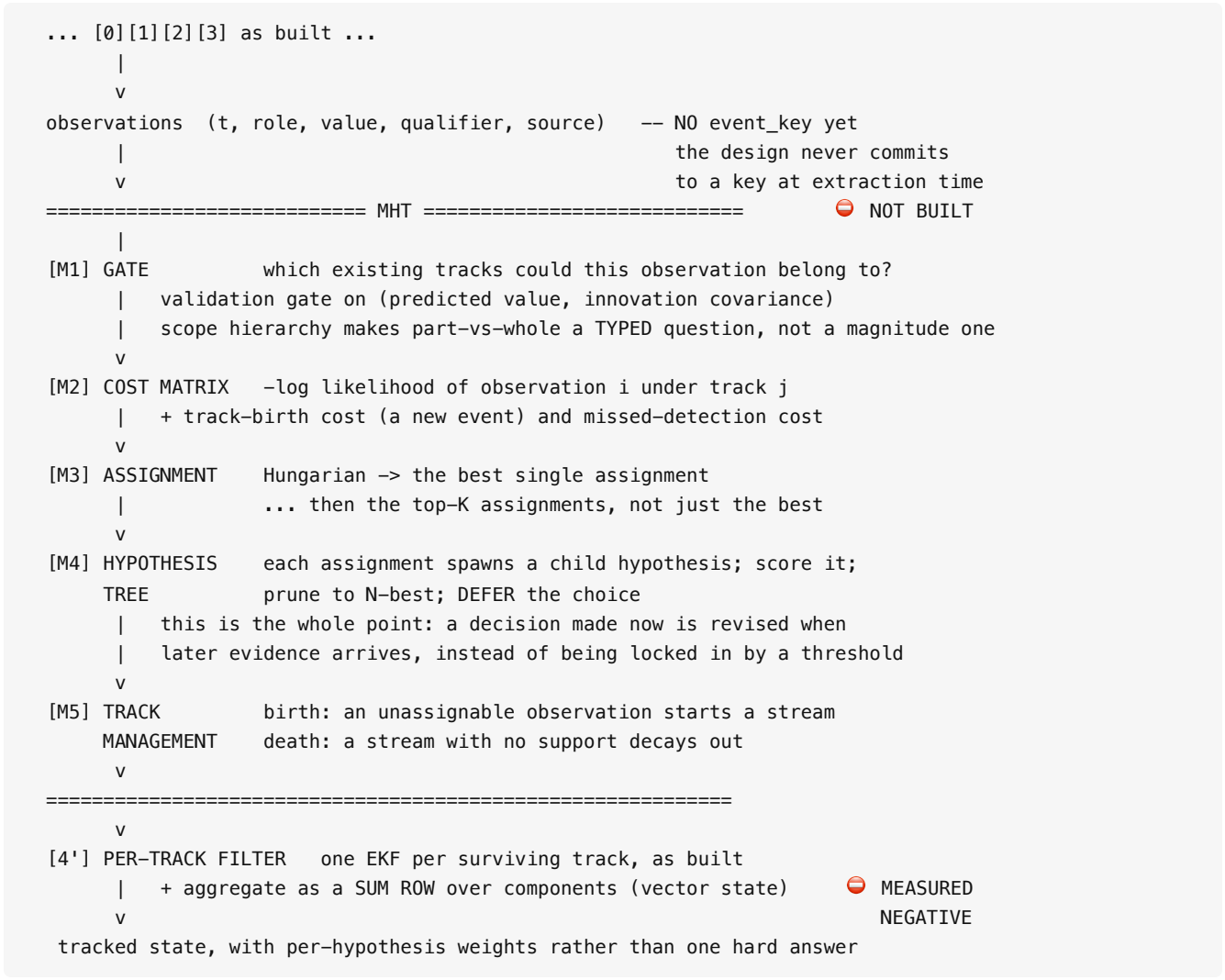
As-built notes

- ⚠ **[3c] is the research blocker**, not [4]. Proximity, GPE tags, record-location and admin rollup have all been tried; the scope gate at [3e] is the first thing that moved it.
- **[3e] is where MHT would have gone.** What ships is hard assignment plus a magnitude rule.
- ⛔ **No MHT.** §3 specifies gate -> Hungarian -> top-K hypotheses -> track birth/death. None of it is built. See §4 below for whether it should be.
- **Everything from [3b] down is OFF BY DEFAULT.** `--event-year`, `--rollup`, `--record-mode`, `--associate` all have to be passed. Defaults reproduce the older numbers on purpose.

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3. EKF pipeline — AS DESIGNED

`EKF_MHT_DESIGN.md` §3. Stages 0-3 are the same; everything from association onward differs. The design's claim is *diarization*: track AND assign, jointly, with decisions deferred.


... [0][1][2][3] as built ...
observations (t, role, value, qualifier, source) -- NO event_key yet
the design never commits
to a key at extraction time
MHT
NOT BUILT
[M1] GATE which existing tracks could this observation belong to?
validation gate on (predicted value, innovation covariance)
scope hierarchy makes part-vs-whole a TYPED question, not a magnitude one
[M2] COST MATRIX -log likelihood of observation i under track j
+ track-birth cost (a new event) and missed-detection cost
[M3] ASSIGNMENT Hungarian -> the best single assignment
... then the top-K assignments, not just the best
[M4] HYPOTHESIS TREE each assignment spawns a child hypothesis; score it;
prune to N-best; DEFER the choice
this is the whole point: a decision made now is revised when
later evidence arrives, instead of being locked in by a threshold
[M5] TRACK MANAGEMENT birth: an unassignable observation starts a stream
death: a stream with no support decays out
[4'] PER-TRACK FILTER one EKF per surviving track, as built
+ aggregate as a SUM ROW over components (vector state)
MEASURED
NEGATIVE
tracked state, with per-hypothesis weights rather than one hard answer

Where as-designed and as-built diverge, and what the measurements say

design element	status	evidence
global decode, 3 states (M4′)	✅ **built 2026-08-25, SHIPS**	pooled RMSE in deaths: Helene −29.4%, Türkiye −7.6%, Aegean −78.8% at one setting — `scope_gate.hmm_gate`
cost matrix (M2)	⛔ **not built, never measured**	−log likelihood of observation *i* under track *j*, plus a track-birth and a missed-detection cost. `mht_associate.py` computes the innovation ingredient per observation in time order but never assembles a matrix, and the two costs it omits are the λ / P_D machinery M5 replaced with a hand-swept σ
deferred assignment (M4)	⛔ not built as MHT	headroom **+0.111 nRMSE** once the null hypothesis is priced — see §4. Taken by the global decode above, which needs no hypothesis tree
track birth/death (M5)	✅ built, ⛔ **measured negative**	nRMSE 0.608 best against the magnitude gate's 0.591 — see §4.1
aggregate as sum row	⛔ **do not build**	loses at every density except 80%, worst where predicted to win (§23)
implied-max reference	⛔ **do not build**	nRMSE 2.590 vs 0.591 on Helene (§25.5)
hard key assignment	✅ built	what shipped until 2026-08-25; the scope gate patched its worst failure
Student-t measurement model	✅ built, ⛔ **measured negative**	Helene −1.7, Türkiye +651; retires no threshold
soft PDA association	✅ built, ⛔ **measured negative**	loses to the hard decode on both; hypotheses are nested, so it double-counts
downward-revision state	✅ built, ⛔ **inert**	correct, but the reports never follow a reclassification down
negative supervision (data-side)	✅ built, ⛔ **superseded**	a plausibility ceiling beats it — see §4.2
per-event plausibility ceiling	✅ built	378.809 → 18.287 on the production model — §4.2
scope membership	✅ built	4/6 cross-event at 7.3% FP, no model — 21 filters → 6
span-embedding router	⛔ blocked on the front end	no current model emits trigger+argument spans on wire copy — §4.4

design element	status	evidence
EKF front-end model	🔄 in build	cold-start mmBERT; English trigger→argument 798 → ~39,800 — §4.4
scope hierarchy declared	✅ built	`rollup.json` `hierarchy` block

4. Is the line far enough along for MHT? — measured, and the answer is no

The question is not "is MHT good" but "how much is left on the table for *any* better association". That is measurable without building anything: assign every observation to the scope it actually fits, using ground truth. It is a ceiling, not a method.

```
shipped scope gate            0.591
oracle hard association       0.537
headroom                     +0.055     (9.3% relative)
```

(Superseded 2026-08-25 — and the correction below is the one that mattered. Perfect two-way assignment buys 0.055 because it cannot REJECT; with a reject option the ceiling doubles, and a global decode over {own, aggregate, reject} then takes most of it without any hypothesis tree. Hungarian assignment solves the half worth nothing. See `EKF_MHT_DESIGN.md` §7.7.)

Perfect association buys 0.055. MHT is a hypothesis tree, a cost matrix, Hungarian assignment and track management — a large subsystem — to compete for a 9% residual.

Corrected 2026-08-19: that oracle prices the wrong thing. It is *two-way* — every observation goes to its own place or to Total — so a figure belonging to no Helene scope at all has no correct home, and it scores Katrina's 1,400 as badly as the shipped gate does. The tell was already in the per-place table below and was read as a curiosity: the gate **beats** the perfect oracle on Florida and South Carolina, because the gate can *drop* and the oracle cannot. MHT's track birth/death **is** a null hypothesis, so the two-way oracle never priced the version of MHT worth building. Adding a reject option (`oracle_gate_three_way`, swept over tolerance rather than fixed at one lucky value):

```
tol   kept    per-place mean
```

2.00 100 0.533 1.00 100 0.533 0.50 85 0.480 0.25 76 0.499

```
shipped scope gate            0.591
three-way oracle              0.480
corrected headroom           +0.111     (18.8% relative)
```

Still a ceiling that uses ground truth, still one event, and the tolerance is tuned and non-monotone — 0.25 is worse than 0.50, so there is no plateau to hide behind. But the prize for association is **double** what MHT was rejected on, and it splits almost evenly (nRMSE): 0.591 → 0.537 is reassignment (+0.055, mostly

Tennessee) and 0.537 → 0.480 is the reject option (+0.057). **About half the prize needs a null hypothesis; the other half does not.**

The per-place breakdown says something sharper:

	shipped gate	oracle
Florida	**0.704**	0.734
Georgia	0.553	0.553
South Carolina	**0.365**	0.558
North Carolina	0.518	0.518
Tennessee	0.817	**0.320**

The gate already **beats** a perfect two-way assignment on Florida and South Carolina, because it has a third option the oracle lacks: *drop*. Florida's 300 and North Carolina's 1400 are not misassigned: 300 is not a casualty figure and 1400 is **Hurricane Katrina's** toll quoted inside a Helene article. Neither belongs to any Helene scope, so no association scheme can place them correctly.

Tennessee is the one genuine association gap (0.817 vs 0.320), and it is instructive: its contaminants are 32, 32, 32, 36, 50 against a truth of 18 — **too large for the state, too small to look national**, so a magnitude test cannot catch them. That is the case for richer association, and it is worth 0.055 across the event.

And 0.537 of the 0.591 is not association error at all. Context audit of all 106 observations puts that residual at cross-event contamination (4.7%), non-casualty numbers (3.8%), a 9.4% unclear tail, and the filter itself — *not* at starvation. See the conclusion below; the recall claim that used to sit here was stale.

Conclusion

Not yet, and MHT is probably not what is missing. In order:

1. ~~**Extraction recall**~~ — **already solved, and the claim behind this was stale.** `extract_long` took the Helene feed from 25 dead observations to 106 (4.2x) and article coverage to 44 of the 45 articles that contain a casualty-bearing sentence. The feed has ~86 sentences carrying both casualty language and a digit, so at 106 the pipeline over-extracts rather than starves. The "25 observations from 70 articles" figure quoted here and in `EKF_MHT_DESIGN.md` §6.2 came from the superseded `tracked_lead` run.
2. **Cross-event contamination — the real top item.** Context audit of all 106 observations: 82.1% are genuine Helene casualties, **4.7% belong to another event** (Katrina 1400, Typhoon 250, Milton 230, Bosnia 16, Mexico's Hurricane John 2), 3.8% are non-casualty numbers (mph, inches), 9.4% unclear. Cross-event carries the *large* values, so it does the most damage per instance.
3. **Non-casualty number rejection** — smaller and easier: speeds, durations, rainfall. Pure extraction typing; no tracker can help.
4. **Multi-source feeds** — the one thing that would make a filter benchmark meaningful, and the one regime where MHT's deferred assignment genuinely earns its keep, because *sources disagreeing about the same event* is real association ambiguity in a way that one wire service's copy is not.

MHT becomes the right move at step 3, not before — with two amendments. Its prize is 18.8% rather than 9%, and the cheapest piece of it has since been built and lost (§4.1). The "starved" clause in the earlier version of this sentence was also stale: the audit above shows the pipeline over-extracts rather than starves.

4.1 M5 track birth: built, and it loses to the fixed ratio it replaces

`tools/ekf_showcase/mht_associate.py`. Tracks advance jointly in time order; each observation is tested by normalized innovation against its candidate tracks, and one that gates out of all of them is born into its own and leaves these streams. No ground truth anywhere.

```
arm                                      Total   per-place   (sigma 4.0)
no gate                                  0.402       5.247
symmetric birth, own+aggregate           0.555       1.059   q_rel 0.20 (the filter's)
symmetric birth, tuned                   2.115       0.636   q_rel 2.00
one-sided birth, tuned                   2.115       0.608   q_rel 2.00
aggregate-only reference                 0.387       0.624   q_rel 0.20
SHIPPED magnitude scope gate             0.316       0.591
three-way oracle (ground truth)          0.308       0.480
```

Read the pair, and the pair is what makes this an unambiguous loss. The two tuned arms buy their per-place improvement by dumping junk into the aggregate: Total nRMSE 2.115 against the gate's 0.316, a 6.7x degradation of the one stream this project calls its honest measurement. That is precisely the failure the shipped gate's three-outcome design exists to prevent — `gate()`'s docstring records that an earlier two-way version rerouted every reject to `__aggregate__` and destroyed the national stream. The associator reproduced a solved bug. So "nRMSE 0.608 against 0.591" understates it; the honest comparison is 2.115/0.608 against 0.316/0.591.

Two causes, both measured:

1. **Judging a stream against its own track is circular.** Every contaminant the track accepts moves the reference the next test uses. Removing the self-reference is worth more than every other knob combined — nRMSE 1.059 → 0.624 at native dynamics. This is the same failure the implied-maximum reference hit on Türkiye, where Turkey was judged against a reference Turkey itself defines, so it is now **two independent mechanisms defeated by one cause**.
2. **The innovation is not informative about scope on a rising toll.** At the filter's own `q_rel = 0.20` the tracks are far too tight to admit real growth — Georgia keeps only `[2, 3]` against a truth peak of 34 — and the sweep must reach `q_rel = 2.00` before real rises survive, by which point only 1–4 observations are ever born. Birth is never the lever; the rerouting is.

This strengthens rather than kills the case for M4. Deferred assignment is the one piece that addresses cause (1) directly: hard assignment commits early and poisons its own reference, which is exactly what keeping rival hypotheses alive exists to prevent. Before this run that was a design preference; it is now the mechanism a measurement implicates.

> **Superseded 2026-08-25.** The paragraph above is the conclusion as it stood on 2026-08-19 and is kept because its diagnosis holds — only its remedy is out of date. Six days later the global decode over {own,

> aggregate, reject} (**M4′**, `scope_gate.hmm_gate`) was built and shipped, and it takes most of the +0.111 **without a hypothesis tree** (Helene −29.4%, Türkiye −7.6%, Aegean −78.8% pooled RMSE in deaths). The headroom splits almost evenly — 0.591 → 0.537 reassignment (+0.055) and 0.537 → 0.480 the reject option (+0.057) — and a tree buys the reassignment half, the one §4 calls the half worth nothing. Cause (1) is still real; deferring the decision is simply no longer the cheapest way to stop it. See the status table in §3 for the current position: *M4 — not built as MHT, taken by the global decode above.*

4.2 The data-side route: built, and beaten by a threshold

The other route to cross-event contamination is negative supervision — withhold an interfering event's records while keeping its text, so its figures become negatives for the same queries. `casualty_loc_muted`, two arms trained identically for four epochs.

It works as a treatment. Blind-test precision up and recall down (0.8119/0.8182 → 0.8273/0.7754), and on Helene it removes 15 of the control's 20 large false positives, cutting ungated per-place error 46.844 → 19.822.

Then a declared per-event plausibility ceiling beats it. Auditing what those large values actually are found that none is a Helene death toll: Asheville's population (94,000), Boone's (19,000), FEMA flood-insurance *policies* (129,933), wellness checks (15,000), power crews (8,000), active-duty troops (1,500), churches (1,100), and two years read as tolls (1,916, 2,004). Only Katrina's 1,400 and Maria's 3,000 are genuinely cross-event.

Ungated per-place mean **nRMSE**, ceiling swept:

ceiling	production	control 4ep	muted
off	378.809	46.844	19.822
20000	18.287	26.961	19.822
2000	18.190	5.853	6.194
1000	18.190	5.057	6.194

At a ceiling of 2,000 — nine times Helene's true toll — **the control beats the muted arm both ungated (nRMSE 5.853 vs 6.194) and gated (3.336 vs 3.729)**, while carrying 81 *more* observations. The ceiling removes only junk; muting removed genuine signal too. Dropping the single 94,000 is worth 20× on the production model.

Verdict: superseded. The arm's pre-registered guard passes only against an undefended control. The suppression is real and learned; it is not worth having.

And neither mechanism touches cross-event. Both true cross-event tolls survive muting and the ceiling alike — at 2,000 Katrina's 1,400 is a plausible magnitude, and a ceiling low enough to catch it is the magnitude gate again. The same holds for the troops and crews: living people in the affected area, wrong for a reason no ceiling sees and no entity-type check reaches. That residue — living people miscounted as dead, and another storm's dead — is what still needs the model to represent who a figure is *about*.

4.3 The Helene reference is a cached artifact and cannot be regenerated

Everything in §4, §4.1 and §4.2 above that cites 5.247 / 0.591 / 0.537 / 0.480 reads one cached file written 2026-08-10, which **no committed state of the repository reproduces**: the `--rollup` flag did not exist in any commit before it was written, and the rollup file was not in the tree either. Comparisons among those figures stand — one frozen artifact — but no new model can be placed on their scale, which is why §4.2 uses a fresh baseline. Full evidence in `../ekf_showcase/muting_arm_results/PROVENANCE.md`; `run_pipeline.py` now records its full invocation and a `-dirty` git marker in every output.

4.4 The front end was rebuilt, and it worked -- but the router's input is still blocked

The association work above all assumes an extractor that produces, per event, a trigger and arguments bound to *that* trigger. Measured 2026-08-20, nothing in the line does.

Span architecture emits a *bag* of triggers and labels every one with the same role. No threshold works: at 0.4 and below the Katrina block is `[95:119]`, the bare name, missing the 1,400 it should bind, while the Helene block runs `[0:212]` and swallows Katrina; at 0.5+ Katrina yields nothing.

Boundary mmBERT `137k-clean` yields nothing at trigger/argument threshold 0.3 and above on English disaster text, and nonsense at 0.1 — trigger `"remote"`, with both `dead` and `location` bound to `"Helene decimated"`. A textbook earthquake sentence returns empty even at 0.1, despite the model scoring trigger 0.710 / argument 0.506 on its own test set.

> **Corrected 2026-08-23.** Two things here were wrong. (a) 0.710 / 0.506 are **relaxed** numbers on the model's own test set; like-for-like on the shared blind test at a pinned 0.5 the base is strict 0.7487 / 0.0913, fair 0.7523 / 0.4939. (b) The "no events above 0.3, nonsense at 0.1" reading came from a harness whose threshold sweep was inert — it set a Schema value the boundary decode never reads, so every row ran at 0.5. Re-measured properly the base forms usable events on 0.0 / 0.0 / 8.3 / 20.0 / 65.0% of Helene windows across 0.5→0.1. The qualitative premise stands; "~0 at every threshold" did not.

The cause is the mix, and it is arithmetic. Counting only corpora that bind arguments to a trigger — DocEE, ChFinAnn and DocFEE do not; they are stored as `entities` + `classifications`:

	English	Chinese	English share
`137k-clean` as built	**798**	20,884	**3.7%**
every available corpus	39,783	20,884	65.6%

The English side is CASIE alone. MAVEN and Mendeley are trigger-only. So an argument F1 of 0.506 is very nearly a Chinese-only number, and English trigger→argument rests on 798 examples — which no threshold reaches.

The rebuild was run, and the arithmetic below was right. 50× more English trigger→argument supervision produced a model that beats the base on all eight held-out heads AND binds the correct death toll 67-100% of the time (binding precision) on Helene wire copy, against the base's 0-7.7%. A pre-

registered form gate initially said the reverse -- it counts firings, not correct ones, and the base fires on 39 of 60 windows while getting 3 right. What still blocks the router is not the mix but the decode: one event instance per type, all spans pooled. See EKF_MHT_DESIGN §7.6.

`tools/train/config/casualty/ekf-frontend-mmbert.yaml` is the cold-start rebuild: 189,284 records, a 50× increase in English trigger→argument, the Chinese corpora kept because they are why the argument head works at all. Split gate clean at 180,660 / 11,486 / 20,571.

The risk is in the config, stated before spending: 72% of the new English trigger+argument data is synthetic, against 20% human-annotated real news, on a line whose recurring failure is in-domain-good / real-news-zero. The gates are on AP prose for that reason.

Smoke, 1× A100-SXM4-40GB: 18.4 samples/s — 34% faster than the extrapolation every earlier cost estimate used. `num_workers` is *not* the bottleneck (18.4 at 0 workers, 18.5 at 4); utilisation swings 28–81% on variable sequence length while memory stays flat at 10.6 GB of 40.

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5. Status & Training Plan on Top of `eb22g` (Updated 2026-11-05)

5.1 What `eb22g` Changes for the EKF Pipeline

In §4.4 above (written 2026-08-23), the EKF front-end was blocked on one architectural limitation: "*What still blocks the router is not the mix but the decode: one event instance per type, all spans pooled.*" Between September and November 2026 (`PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`), that exact limitation and the three training starvations behind it were resolved in `eb22g` :

1. **Multi-instance individuation (`event_records: true`):** Event instances are now seeded per trigger anchor span (`record_instance_queries = 32`), not pooled 1-per-type.
2. **Relational trigger-argument binding (`record_link_mode: junction`):** Gated bilinear + geometry scoring with contrastive column loss (`junction AUC 0.74–0.77`) stops false or competing triggers from stealing figures.
3. **Within-document event & entity coreference (`TriggerLink` + `EntityCorefLink`):** Multi-mention events within an article consolidate into a single cluster (`pair AUC 0.802–0.843`) while distinct same-type events stay split.
4. **Balanced positive-negative supervision (`pair_split_reduction: true` , split `_list_field_bce`):** Lifted 64-doc overfit `@0.5` argument strict micro-F1 to `0.4319` on CASIE and `0.8005` on CMNEE, lifted full-mix Epoch 5 validation `event_argument` strict micro-F1 at `@0.5` to **`0.3038`** (`0.3046` Ep 4; relaxed **`0.5342`** , fair **`0.5221`**) and `event_trigger` to **`0.5720`** (recall **`0.7399`** , fair **`0.5980`**), and on the **20,602-document blind test** (per-head thresholds) reached **`0.4637`** strict / **`0.5985`** relaxed / **`0.5901`** fair `event_argument` micro-F1 and **`0.6994`** strict (`0.7227` `@0.9`) / **`0.7167`** relaxed `event_trigger` micro-F1.

5.2 Audit of Outdated Data & Configs in `datasets/` and `data/`

Auditing `datasets/` and `tools/train/config/{casualty,gate}/` against `eb22g` surfaced four concrete mismatches that must be fixed before training the `eb22g` EKF models:

1. **Anchor mismatch between `data/casualty_ml` and `run_pipeline.py` (`build_casualty_schema`):** - `data/casualty_ml.{train,val,test}.jsonl` (62,703 real EN/TR/ZH news records) stamps `record_metadata: {"casualty_report": {"mode": "natural", "anchor": "location"}}`. - Meanwhile, `tools/ekf_showcase/run_pipeline.py:345` (`build_casualty_schema`) constructs the inference schema with **`anchor="dead"`**! On the boundary record head in `natural` mode, the anchor field (`role_index 0`) seeds the record instances: training with `anchor: "location"` and decoding with `anchor: "dead"` (or decoding an injured-only report with `anchor: "dead"`) mismatches the trained anchor. - **Fix:** Standardize the anchor between training and `build_casualty_schema` (either `anchor: "location"` for place-keyed `casualty_report` structures, or event-trigger-anchored `casualty_events` where the event trigger is always the anchor and `location`, `dead`, `injured`, `missing`, `displaced`, `rescued` are argument roles).
2. **Label-map version mismatch (`labels/unified.yaml` vs `labels/unified-full-v4.yaml`):** - Existing configs (`casualty-full-multilingual.yaml`, `casualty-events-boundary.yaml`, `gate3-mmbert.yaml`) reference `labels/unified.yaml` (or no `labels_file`) and warm-start from `eb16-rebuild-tr` (which also used the deprecated charclass length proxy). - **Fix:** Every model warm-started from `whr778/gliner2-eb22g` must reference `labels_file: labels/unified-full-v4.yaml` and inherit `eb22g`'s structural boundary settings (`enable_rotary_endpoints: false`, `coref_link_distance: mlp`, etc.).
3. **Zero-replay forgetting in `casualty-full-multilingual.yaml`:** - Training `casualty-multilingual-eb16tr` on `data/casualty_ml` alone (0% replay) collapsed general capability (`entity 0.5695 -> 0.0801`, `event_type 0.7545 -> 0.0209`). - **Fix:** Include **30% exact replay** drawn from `eb22g`'s training pool (`build_warmstart_mix.py` / `build_137k_replay.py` updated for `eb22g`).
4. **Stale cached observation files in `datasets/`:** - `datasets/helene2024/_rescore/`, `datasets/turkey2023/_rescore/`, and `datasets/ekf_showcase/*.json` hold observations extracted in August/September 2026 (before `build_helene_feed.py` stripped embedded AP related-coverage rails on 2026-09-21, and before `eb22g`). - **Fix:** Once the `eb22g` EKF models are trained, regenerate the feeds (`build_helene_feed.py`, `build_turkey_feed*.py`, `build_aegean_feed.py`) and re-extract all four real-event benchmark feeds (`helene2024`, `turkey2023`, `aegean2020`, and the double-blind `venezuela_2026` holdout).

5.3 Step-by-Step Training Plan for the EKF Pipeline Components (Post-`eb22g`)

With `eb22g` (`bcfe500c`, `whr778/gliner2-eb22g`) complete and verified on its 20,602-record blind test:

Step	Component / Config	Base Checkpoint	Training Data & Recipe	Pre-Registered Acceptance Gate
1	**Zero-Shot `eb22g` Front-End Readout** (no training, ~$0)	`whr778/gliner2-eb22g` (`best`)	Run `frontend_gates.py` + `binding_accuracy.py` on the Helene/Katrina multi-event passages and DocEE 27-label set	Does `eb22g` pass **Gate 2** (separating Helene and Katrina into two distinct event instances in one passage) and bind the right toll?
2	**Stage 0 Relevance & Normalizer Gate (`gate4-eb22g.yaml`)**	`whr778/gliner2-eb22g`	`data/gate3` (token-balanced `en` / `tr` / `zh` `relevance` + `toll_kind`) + `qualifier` / `source` normalizer tasks + 30% `eb22g` replay; `labels/unified-full-v4.yaml` ; 4 epochs	Val-calibrated threshold (`sweep_thresholds`): EN/TR/ZH admission >= `gate3-mmbert` (`97%` TR, `98%` ZH), 0 regression on Helene/Aegean/Türkiye-EN
3	**Stage 2 Multilingual Casualty Extractor (`casualty-ml-eb22g.yaml`)**	`whr778/gliner2-eb22g`	`data/casualty_ml` (62,703 real EN/TR/ZH news records, anchor aligned with `build_casualty_schema`) + `data/casualty_events` + **30% exact `eb22g` replay**; `pair_split_reduction: true` , `record_link_mode: junction` ; 4 epochs	(1) `casualty_ml` blind-test `structure` strict micro-F1 > `0.6078` across all 3 languages; (2) `eb22g` replay heads hold within `±0.02` ; (3) Helene binding precision >= `67–100%` and Turkish digit-in-location < `5.8%`
4	**End-to-End Re-Extraction & Re-Baseline of `datasets/`**	`gate4-eb22g` + `casualty-ml-eb22g`	Re-run `run_pipeline.py` (`--window long --associate record --scope-filter --scope-method decode`) on `helene2024` , `turkey2023` (`en` / `tr` / `zh`), and `aegean2020` , then score `venezuela_2026` once	Pooled RMSE in deaths <= current `hmm_gate` baselines (`20.7` Helene, `10,695.5` Türkiye, `15.7` Aegean) and beats `est_last_value` on all feeds

5.4 Are We Ready for MHT? — The Exact Prerequisite Gate

Answer: Not before Steps 1–4 above complete — and Steps 1–4 will tell us whether MHT has any remaining headroom to win.

Specifically: 1. **Why MHT was not justified previously (§4):** On single-event-per-region feeds (Helene, Türkiye–Syria, Aegean) where observations are already keyed by `(event_type, location)` , the two-way assignment oracle had **zero** headroom over the shipped gate (`29.3` vs `29.3` deaths on Helene), and the 3-state global Viterbi decode (`hmm_gate` over `{own, aggregate, reject}`) captured 73% of the 3-way reject oracle's headroom without a hypothesis tree. 2. **Where MHT *can* earn its keep once `eb22g` is in place:** MHT (`[M1–M4]` : cost matrix over innovation likelihood + `candidate_states` / `TriggerLink` embedding affinity → top- `K` hypothesis tree) is needed **if and only if** a feed contains **multiple concurrent**

events of the same type in the same geographic scope (e.g., two simultaneous attacks or accidents in the same city/province, or an unkeyed stream where `location` is absent on >20% of observations so observations must be associated to tracks by `candidate_states` embedding + trajectory innovation jointly). 3. **The MHT Go/No-Go Gate (run immediately after Step 4):** - Re-run `reject_headroom.py` and the two-way vs three-way oracle on the Step-4 `casualty-ml-eb22g` observations across `helene2024`, `turkey2023`, `aegean2020`, and the multi-incident synthetic feed (`datasets/ekf_showcase/feed_multi.jsonl`). - **Go criterion for MHT:** If the gap between `hmm_gate` (hard key + Viterbi `{own, aggregate, reject}`) and the oracle with ground-truth track assignment exceeds **15% relative pooled RMSE** on multi-incident feeds (and Gate 2 passes so each incident carries distinct `candidate_states` embeddings), we build `[M1–M4]` using the `TriggerLink` / `candidate_states` cosine + Kalman innovation cost matrix. Otherwise, `hmm_gate` remains the complete association layer for Paper 1.